A new specialized PG degree program in the Department of Mechanical Engineering

M.Tech. in Advanced Manufacturing Engineering

Program Commencing from July 2026



Department of Mechanical Engineering
Indian Institute of Technology Jodhpur
2026

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M.Tech. in Advanced Manufacturing Engineering

Concept Note

Background

Manufacturing is an important sector for India's economic growth, technological self-reliance, and industrial competitiveness. With the rapid development of the manufacturing sector, there is a growing demand for highly skilled postgraduates with strong fundamentals and exposure to emerging manufacturing technologies. The proposed M.Tech. in Manufacturing Engineering is designed to address this need by providing advanced theoretical knowledge, practical insights, and research-oriented training in advanced manufacturing engineering. The program aims to develop professionals capable of understanding, designing, analyzing, and optimizing manufacturing processes across conventional and advanced domains. Emphasis is placed on materials processing, evaluation, and design, advanced and non-traditional manufacturing techniques, welding and joining technologies, additive manufacturing, precision and micro-manufacturing, vacuum technology, and non-destructive testing. The curriculum integrates core courses that strengthen fundamental concepts with flexible electives that allow students to specialize in line with industry trends and research interests.

The inclusion of electives such as Additive Manufacturing, MEMS and Microsystems Technology, Precision and Micro-Manufacturing, Surface Engineering, Vacuum Technology in Manufacturing, Artificial Intelligence for manufacturing applications and NDT in Manufacturing ensures alignment with Industry 4.0, aerospace, defense, electronics, energy, and advanced materials sectors. Additional electives in casting, composites, quality control, and design methodology further enhance the innovation capabilities. A significant component of the program is the two-semester thesis, which provides students with an opportunity to undertake in-depth research, product development, or industry-relevant problem solving. This Program will prepare students for careers in R&D organizations, advanced manufacturing industries, start-ups, and doctoral research programs.

In summary, the M.Tech. in Manufacturing Engineering is structured to produce technically competent, innovation-driven, and research-ready engineers/researchers who can contribute effectively to national strategic initiatives such as *Make in India*, *AtmanirbharBharat*,

and the global manufacturing ecosystem.

Need of the program

- To develop industry-ready, highly skilled manufacturing professionals aligned with current and emerging industrial needs
- To strengthen the research ecosystem in advanced manufacturing technologies
- To support national self-reliance initiatives such as Make in India, National Manufacturing Mission, and Atmanirbhar Bharat while contributing to global manufacturing competitiveness
- To advance research and innovation in strategic sectors including defense, space, semiconductor manufacturing, etc.
- To adopt in response to the changing geopolitical landscape and need for self-reliance

Objectives of the Program

- To develop highly skilled professionals capable of understanding, analyzing, and optimizing the processes across both conventional and advanced manufacturing domains through innovation and research-driven approaches
- To strengthen foundational concepts in Manufacturing Engineering while offering flexible elective pathways that enable specialization aligned with emerging industry trends and advanced research domains.
- To nurture manufacturing engineers capable of driving innovation, technology-based entrepreneurship, and translational research in emerging manufacturing fields.
- To prepare innovation-driven manufacturing engineers with strong technical and research competencies to support national self-reliance initiatives such as Make in India, National Manufacturing Mission and Atmanirbhar Bharat.
- To develop expertise in sustainable manufacturing aligned with the United Nations SDGs, particularly, Affordable and Clean Energy (SDG 7), Industry, Innovation and Infrastructure (SDG 9), and Responsible Consumption and Production (SDG 12).

Graduate Attributes

- Ability to undertake independent academic and industrial research in advanced manufacturing using analytical, computational, and experimental methodologies
- Strong conceptual and applied knowledge in emerging areas such as hybrid

manufacturing, MEMS, precision and micro-manufacturing, and additive manufacturing

- Ability to adopt a multidisciplinary approach to analyze and solve complex manufacturing problems
- Capability to apply critical, creative, and innovation-driven thinking in addressing engineering challenges
- Proficiency in articulating technical ideas effectively through written reports, publications, and oral presentations
- Demonstration of professional integrity, ethical responsibility, and commitment to societal and sustainable development goals

Learning Outcomes

- Demonstrate in-depth understanding of fundamental principles governing conventional and advanced manufacturing processes
- Knowledge of engineering materials, manufacturing processes and systems in the context of product development
- Analyze, design, and optimize manufacturing processes using analytical, computational, and experimental tools
- Integrate multidisciplinary expertise to develop intelligent manufacturing solutions and translate them into scalable, technology-based entrepreneurial ventures

Eligible Branches and Justification

Regular students: B.Tech., B.E. (Mechanical Engineering, Production Engineering, Industrial Engineering, Production and Industrial Engineering, Aerospace Engineering, Mechatronics Engineering, Applied Mechanics, Automobile Engineering, Aeronautical Engineering, Power plant engineering, Nanotechnology, Materials Engineering, Instrumentation Engineering). Valid GATE score in Mechanical Engineering/Aerospace Engineering/ Production and Industrial Engineering, or from any of the CFTIs with CPI or CGPA 8/10 or above.

Sponsored students: B.Tech., B.E. (Mechanical Engineering, Production Engineering, Industrial Engineering, Production and Industrial Engineering, Aerospace Engineering, Mechatronics Engineering, Applied Mechanics, Automobile Engineering, Aeronautical Engineering, Power plant engineering, Nanotechnology, Materials Engineering, Instrumentation Engineering)

Student Intake Number and Strategy:

A maximum of 15 students is planned to be admitted to this program. These include 8 GATE-qualified students (MoE fellowship), 7 self-sponsored students.

Course Structure

Semester-wise Distribution of Credits

Cat	Course Code	Course Title	L-T-P	Credits	Cat	Course Code	Course Title	L-T-P	Credits
I Semester					**II Semester**				
C	MEL7XXX	Engineering Mathematics	3-0-0	3	C	MEL7XXX	Artificial Intelligence for Manufacturing Applications	3-0-0	3
C	MEL7XXX	Materials Design and Evaluation Techniques	3-0-0	3	C	MEL7XXX	Conventional and Advanced Manufacturing Processes	3-0-0	3
C	MEL7XXX	Advanced Material Processing	3-0-0	3	E	MEL7XXX	Elective II	3-0-0	3
E	MEL7XXX	Elective I	3-0-0	3	E	MEL7XXX	Elective III	3-0-0	3
NG	LAN7XX0	Technical Communication	1-0-0	S/X	NG	OAL7XX0	Innovation and IP Management	1-0-0	S/X
Total				**12**	Total				**12**
					T	MES7010	Summer Thesis		**8**
III Semester					**IV Semester**				
T	MET7XX0	Thesis	-	16	T	MET7XX0	Thesis	-	16
Total				**16**				Total	**16**

Program Compulsory Courses:

Semester	Course Code	Course Title	L-T-P	Credits
I	MEL7XX0	Engineering Mathematics	3-0-0	3
	MEL7XX0	Materials Design and Evaluation Techniques	3-0-0	3

	MEL7XX0	Advanced Material Processing	3-0-0	3
II	MEL7XX0	Artificial Intelligence for manufacturing applications	3-0-0	3
	MEL7XX0	Conventional & Advanced Manufacturing Processes	3-0-0	3

Program Elective Courses:

S. No.	Course Code	Course Title	L-T-P	Credits
1	MEL7XX0	MEMS and Microsystems Technology	3-0-0	3
2	MEL7XX0	Additive Manufacturing	3-0-0	3
3	MEL7XX0	Theory of Arc Welding	3-0-0	3
4	MEL7XX0	Precision and Micro-Manufacturing	3-0-0	3
5	MEL7XX0	Engineering Design Innovation	3-0-0	3
6	MEL7XX0	Vacuum Technology in Manufacturing	3-0-0	3
7	MEL7XX0	NDT for Manufacturing Applications	3-0-0	3
8	MEL7XX0	Computer-Aided Inspection and Quality Control	3-0-0	3
9	MEL7XX0	Welding: Metallurgy, Defects, and Testing	3-0-0	3

Program's Structure

S. No.	Course Type	Cat.	Credits
1	Program Compulsory	C	15
2	Program Electives	E	9
3	M.Tech Summer Thesis	T	8
3	M. Tech. Thesis I	T	16
4	M. Tech. Thesis II	T	16

Total Grade			**64**
5	Non-Graded Courses	NG	0
Total Grade			**64**

Course content of Program Compulsory and Elective Courses

Title	**Materials Design and Evaluation Techniques**	Number	ME7XX
Department	Mechanical Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M. Tech. Advanced Manufacturing Engineering	Type	C
Prerequisite	Nil		

Objectives

The course aims at imparting knowledge related to:

1. Mechanical behaviour of materials and developing systematic procedure for selecting materials and processes.
2. Non-destructive evaluation of materials and components.

Learning Outcomes

The students are expected to:

1. understand the mechanical behaviour of materials and the role of materials in engineering design.
2. understand with the various non-destructive evaluation techniques for determining component integrity

Contents:

1. Classification of Materials, Relationship between structure, processing and property of material, Types of tests for determining the mechanical behavior of materials: Tension Test, Compression Test, Bending Test, Torsion Test, hardness, fatigue, creep, impact, and their role in materials processing (11 Lectures)
2. Theories of Failure (4 Lectures)
3. Materials Selection and Design (8 Lectures):
    a. Materials in design, The Design Process, Types of Design, Design Tools and Materials Data
    b. Engineering Materials and their properties – Families of Engineering materials, Materials Information for design, Material Properties and their units
    c. Material Property Charts, Material Selection Strategy, Translation, screening, Ranking, Documentation, Attribute limits and Material Indices, The selection Procedure, Computer Aided Selection, Case studies
4. Process Selection (6 Lectures): Classifying Processes, The processes: shaping joining, finishing. Process-Property Trajectories, Processes Selection and cost: The Strategy, Selection Matrices, Limitations, Ranking: Process cost, Computer aided Process selection.
5. Non Destructive Testing (10 Lectures): Visual Inspection and Liquid Penetrant Inspection, Eddy Current Testing, Magnetic Particle Inspection, Ultrasonic Inspection, Radiography Inspection, Neutron Radiography, Case studies.

Text Books

1. Dieter G E , Mechanical Metallurgy, McGraw Hill
2. Ashby M F, Materials Selection in Mechanical Design, Elsevier

Reference Book

1. ASM Handbook, Volume 17: Nondestructive Evaluation of Material, **Editor:** Aquil Ahmad and Leonard J. Bond

Online Course Material:

1. https://archive.nptel.ac.in/courses/112/107/112107259/
2. https://archive.nptel.ac.in/courses/112/104/112104122/

Title	**Engineering Mathematics**	Number	MEL7XX
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M. Tech. and Ph.D.	Type	C

Objectives:

1. Equip students with a comprehensive understanding of advanced mathematical tools (vector calculus, linear algebra, differential equations) and their application in modeling and solving complex physical and engineering systems using ordinary and partial differential equations.
2. Introduce foundational concepts in probability and statistics to enable students to interpret data and analyze uncertainty in engineering processes.

Learning Outcomes

1. Students will proficiently apply vector calculus, tensor analysis, matrix operations, and both analytical and numerical methods for solving ODEs and PDEs to address engineering problems involving multivariable systems, transformations, and physical contexts like vibrations and heat transfer.
2. Students will effectively use statistical methods to analyze experimental data, conduct hypothesis testing, and develop simple predictive models for engineering applications.

Content

1. *Vectors & Tensors* - Vector Algebra; Functions of several variables; Curves and surfaces; Multiple integrals; Gauss (divergence), Stokes and Green's theorems; Tensor notation, transformation and operations. **[9 lectures]**
2. *Linear Algebra* - Matrix operations; LU factorization; Reduced row echelon form and rank; Independence, basis & dimension; Orthogonality, projection and least square approximation; Basis Vectors; Determinant properties; condition number; Eigenvalues and Eigenvectors - including examples; Singular Value decomposition. **[8 lectures]**
3. *ODEs* - First order ODEs; second order linear ODEs; system of ODEs; phase-plane relationship; series solutions of ODEs; Laplace transforms; Fourier series; Sturm-Liouville problems; Fourier transforms; Examples - mass-spring-damper system, multi-degree of freedom discrete vibration systems. **[8 Lectures]**
4. *Numerical ODE* - Initial value problem using Euler's Method, Runge-Kutta Method, Integrating Higher order ODE using system of ODEs, stiff problems, Boundary value problem using shooting method. **[4 Lectures]**
5. *PDEs* - Characterization of PDEs, Separation of Variables, Examples - Unsteady heat conduction equation, string vibrations - Eigen modes etc, Cauchy equilibrium equation-applied on thin elastic plate. **[4 Lectures]**
6. *Probability and Statistics* - Experiments, outcomes, events, central limit theorem, random variables, probability distributions, distributions of several random variables, random sampling, point estimation, confidence intervals, Testing hypotheses, Goodness of fit, regression, correlations. Examples from thermofluids - temperature data in Jodhpur, rainfall data. **[6 Lectures]**

Textbook:

1. Kreyszig, E. (2011). Advanced engineering mathematics (10th ed.). John Wiley & Sons.

Reference Books:

1. Strang, Gilbert. Introduction to Linear Algebra. 5th ed., Wellesley-Cambridge Press, 2016.
2. Borisenko, A. I., and I. E. Tarapov. Vector and Tensor Analysis with Applications. Translated by Richard A. Silverman, Dover Publications, 1979.
3. Chapra, S. C., & Canale, R. P. (2021). Numerical methods for engineers (8th ed.). McGraw-Hill Education

Online Course Material: A beyratne, R. (2021). Volume 1. A brief review of some mathematical preliminaries (Version 1.0: 2 Dec 2006; updated 09 Jul 2021) URL: https://web.mit.edu/abeyaratne/lecture_notes.html

Title	**Advanced Material Processing**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M. Tech. in Advanced Manufacturing Engineering	Type	C
Prerequisite	Nil		

Objectives:

The course aims at imparting knowledge related to:

1. To understand processing principles of advanced and emerging materials.
2. To correlate process-structure-property-performance relationships.
3. To explore hybrid methods suitable for high-performance applications

Learning Outcomes:

The students are expected to:

1. Understand the processing principles of advanced and emerging materials.
2. Analyze process-structure-property-performance relationships.
3. Design and evaluate hybrid processing routes for high-performance applications.

Contents:

1. ***Introduction to novel materials (9 Lectures):***
   Classification: Metal matrix composites (MMCs), ceramic matrix composites (CMCs), polymer composites, high-entropy alloys, bulk metallic glasses, functionally graded materials (FGMs), smart/functional materials. Manufacturing challenges: Poor machinability, high reactivity, defects, high melting points.
2. ***Advanced powder-based processing (10 Lectures):***
   Advanced compaction & sintering: Spark Plasma Sintering (SPS), Hot Isostatic Pressing (HIP), Additive Manufacturing (AM) for novel materials: Laser Powder Bed Fusion (LPBF/SLM), Electron Beam Melting (EBM), Directed Energy Deposition (DED), Binder Jetting for ceramics. Process parameters optimization, defect formation (keyhole porosity, lack-of-fusion, residual stresses), in-situ monitoring, and post-processing.
3. ***Advanced solid-state processing (8 Lectures)***
   Severe Plastic Deformation (SPD) for ultrafine-grained materials, Friction Stir Processing/Welding for composites and dissimilar materials, Thermo-mechanical processing of advanced alloys (e.g., superalloys, titanium alloys, high-entropy alloys)
4. ***Non-conventional and hybrid processing techniques (12 Lectures)***
   Hybrid processes: Laser-assisted forming/machining, ultrasonic vibration-assisted processing, electromagnetic forming, Electro-discharge grinding (ECG). Advanced surface treatments: Thermal spraying, ion implantation, sol-gel coatings, and PVD/CVD coatings. Processing of ceramics and composites: Microwave sintering, and chemical vapor infiltration (CVI)

Text Books

1. J.T. Black, R.A. Kohser, DeGarmo's Materials and Processes in Manufacturing, Wiley publication, 2017.
2. T.W. Clyne and P.J. Withers, An Introduction to Metal Matrix Composites, Cambridge University Press, 2003
3. I. Gibson, D. Rosen, B. Stucker, M. Khorasani, Additive Manufacturing Technologies, Springer Cham, 2020

Reference Book:

1. R. M. Jones, Mechanics of Composite Materials, 2nd Edition, CRC Press, 2015.

Online Course Material:

1. Advanced Materials and Processes by Prof. Jayanta Das https://onlinecourses.nptel.ac.in/noc19_mm13/preview
2. Metal Additive Manufacturing, Prof. Janakranjan Ramkumar, https://nptel.ac.in/courses/112104312
3. Fundamentals of Material Processing - Part 2 by Prof. Shashank Shekhar https://onlinecourses.nptel.ac.in/noc22_mm10/preview

Title	**Conventional and Advanced Manufacturing Processes**	Number	ME7XXX
Department	Mechanical Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B. Tech./ M. Tech.	Type	C
Prerequisite	Nil		

Objectives: The course aims at imparting knowledge related to:

1. To develop a fundamental understanding of conventional and advanced manufacturing processes, including their working principles, material removal mechanisms, process capabilities, and limitations.

2. To familiarize students with modern manufacturing practices and emerging technologies, such as Industry 4.0, Industrial IoT, sustainable manufacturing, and AI/ML applications in manufacturing.

Learning Outcomes: The students are expected to:

1. Analyze and select appropriate conventional and advanced manufacturing processes based on working principles, process capabilities, and application requirements.

2. Understand modern manufacturing concepts and emerging technologies, including Industry 4.0, intelligent manufacturing, and sustainable manufacturing practices.

Contents:

Introduction: Importance of manufacturing, historical perspective, classification (1 Lecture).

Conventional manufacturing processes (15 Lectures):

Advances in casting – Casting Terminology, Gating design, shrinkage and riser design, Metal flow and solidification, Chvorinov's rule, novel innovations in casting, Casting defects and inspection of castings.

Advances in welding – Arc Welding Processes, Welding power source and its characteristics, Resistance welding processes, friction stir welding (FSW), Radiant beam welding and hybrid welding processes.

Advances in forming – Mechanical behavior of metals in elastic and plastic deformation, stress-strain relationships, Yield criteria, Fundamentals of metal working, Analysis of bulk forming processes, High energy rate forming: EMF, EHF and explosive forming, cryogenic forming, advanced materials in forming.

Advances in Machining - Basic working principle, Mechanics of Metal Cutting: Geometry of single point cutting tools; Mechanism of chip formation, Analysis of machining forces and power requirement, Finishing and Super-Finishing Processes

Advanced manufacturing processes (15 Lectures): working principles, material removal mechanisms, process capabilities, applications and limitations

Mechanical Material removal – Abrasive jet Machining, Abrasive water jet machining, Ultrasonic Machining.

Thermal Material removal – Electro-discharge machining, Electron beam machining, Laser beam machining, Ion Beam Machining.

Electro chemical machining – Mechanism of material removal and estimation of MRR, Electrochemical grinding and Electro-stream drilling

Modern manufacturing practices (8 Lectures): Introduction to Industry 4.0, Industrial IoT, Computer integrated manufacturing, Green and sustainable manufacturing, ML/AI applications in manufacturing, Manufacturing of bio-materials and bio-inspired manufacturing, Industrial case studies.

Text Books

1. Advanced Machining Processes, V. K. Jain, Allied Publishers
2. Manufacturing Science, Ghosh A., and Mallick A. K., EWP
3. Fundamental of modern manufacturing Materials, Processes and systems, M.P. Groover

Reference Book:

1. Advanced Machining Processes, Hasan El-Hofy, McGraw-Hill

Online Course Material:

1. Advanced Machining Processes, IIT Kanpur Prof. V. K. Jain https://nptel.ac.in/courses/112104028
2. Manufacturing Process Technology - II, Prof. Shantanu Bhattacharya https://nptel.ac.in/courses/112104204

3. Manufacturing Processes I, Prof. Shan, Prof. Gupta, Dr. Pradeep
   https://nptel.ac.in/courses/112107144/#
4. Manufacturing Processes II, Prof. A.B. Chattopadhyay, Prof. A.K. Chattopadhyay, Prof. S. Paul
   https://nptel.ac.in/courses/112105127/

Title	**Artificial Intelligence for Manufacturing Applications**	Number	ME7XX0
Department	Mechanical Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B. Tech./ M. Tech.	Type	C
Prerequisite	Nil		

Objectives

The course aims to:

- To develop the ability to apply ML/AI techniques for modeling, optimization, monitoring, and decision-making in manufacturing systems, enabling data-driven improvement of process performance, quality, and productivity.
- To equip students with the skills to integrate industrial data, physics-based understanding, and intelligent algorithms for building smart manufacturing solutions, including predictive maintenance, in-situ monitoring, and digital twin development.

Learning Outcomes

The students are expected to:

- Apply ML/AI techniques to analyse manufacturing data for process optimization, quality improvement, and intelligent decision-making.
- Develop data-driven models for in-situ monitoring and predictive health management of manufacturing systems.
- Design and evaluate digital twin and smart manufacturing solutions by integrating industrial data with physics-based understanding.

Contents:

Module 1: Industrial Data Acquisition & Smart Manufacturing Systems - Sensors in Manufacturing, Introduction to industry 4.0, Industrial IoT, Smart factories, Data Preprocessing in Industrial Settings, communication protocols (6 Lectures)

Module 2: ML Preliminaries – Supervised & Unsupervised Learning, Neural Networks & Deep Learning Basics, Dimensionality reduction (PCA), Handling noisy industrial data, Model Evaluation & Overfitting Control (7 Lectures)

Module 3: ML/AI approaches for in-situ monitoring of manufacturing processes – Time-Series Learning, Computer Vision in Manufacturing, AI algorithms for Anomaly Detection, Condition Monitoring Fundamentals (7 Lectures)

Module 4: ML/AI approaches for process optimization – Surrogate Modeling, Numerical modelling of manufacturing process, Accelerated process simulations using data driven approaches, Physics informed AI methods, Operator Learning & Surrogate PDE Models, Optimization techniques (10 Lectures)

Module 5: ML/AI approaches for digital twin development and computational design – Digital Twin Architecture, Generative AI in manufacturing, Inverse Design & Optimization (6 Lectures)

Module 6: Industrial case studies (3 Lectures)

Text Books

1. M. Soroush & R. D. Braatz (2024). "*Artificial Intelligence in Manufacturing: Concepts and Methods*," Springer.
2. S. K. Pal , D. Mishra , A. Pal , S. Dutta , D. Chakravarty & Srikanta Pal (2022). *"Digital Twin – Fundamental Concepts to Applications in Advanced Manufacturing,"* Springer Cham.

Reference Books

1. A. Géron (2022). Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow. O'Reilly Media, Inc.

Online self-learning

1. **Introduction to Industry 4.0 and Industrial Internet of Things – https://nptel.ac.in/courses/106105195**
2. **Introduction to Deep Learning – https://www.youtube.com/playlist?list=PLtBw6njQRU-rwp5_7C0oIVt26ZgjG9NI**

	Title	**MEMS & Microsystems Technology**	Number	MEL7XX0
	Department	Mechanical Engineering	L-T-P [E]	3–0–0 [3]
	Offered for	PG	Type	E
	Prerequisite	Nil		

Objectives

The instructor will :

1. cover fundamental understanding of miniaturization and related functional advantages in various engineering disciplines and corresponding systems/devices
2. teach overview/working principles, modeling, analysis, design, manufacture and packaging of Microsystems along with the state of the art awareness of this domain

Learning Outcomes

The students are expected to have :

1. practical and simulation exposure to understand the complexity in design and manufacturing of Microsystems.
2. basic Microsystems based research and training to develop Micro-systems modules for commercial and futuristic applications.

Contents:

1. Introduction : Fundamentals and applications to Microsystems (2 lectures)
2. Engineering Mechanics For Microsystems Design: Static Bending of Thin Plates, Mechanical Vibration, Thermo-mechanics, Thin-Film Mechanics, Stress Analysis Problems (4 lectures)
3. Fluid Mechanics at small scales: Viscosity of Fluids, Basic Equations in Continuum Fluid Dynamics, Incompressible Fluid Flow in Micro-conduits (3 lectures)
4. Heat Conduction in Solids: General Principle of Heat Conduction, Fourier law of Heat Conduction, Newton's Cooling Law, Solid-Fluid Interaction, Heat Conduction in Multilayered Thin Films (3 lectures)
5. Scaling Laws in Miniaturization (3 lectures)
6. Materials for MEMS and Microsystems: Silicon as Substrate Material, Silicon Compounds, Gallium Arsenide, Quartz, Polymers, Packaging Materials, polymer MEMS (4 lectures)
7. Microsystems Fabrication Processes: Photolithography, micro molding/replica molding process, 3D printing for polymer processing, Diffusion, Oxidation, Chemical/Physical Vapor Deposition, Deposition by Epitaxy, Etching, Bulk and Surface Micro manufacturing. (8 lectures)
8. Characterization techniques: Overview of various visualization techniques such as FESEM, TEM, Optical microscope etc. (3 lectures)
9. Working Principles of Microsystems Devices: Working Principles, Design, Manufacturing and Packaging of Micro-sensors, Actuators, Inertial Sensors and Fluidic devices (8 lectures)
10. Microsystems Design Considerations: Selection of Materials, Manufacturing Processes, Signal Transduction, processing with respect to microsystem Packaging, modeling & simulation aspects (4 lectures)

Laboratory Experiments:

Design and development of micro channel/features with soft lithography; Development of micro-features with the help of electrodeposition; Design and fabrication of micro system device; Design and testing of micro fluidics device through diffusion and capillary action; Silicon wafer handling, RCA cleaning and surface effect analysis with plasma treatment; Modeling and analysis of MEMS based Device such as Capacitive Pressure Sensor, Micro-Inertial Sensors, Micro actuators, Studies based on Fluid Structure Interaction (FSI); Simulation studies on Wet Etching in Anisotropic condition, Meshing and Analysis based on Process Simulation Result, Coupled Package-Device Modeling

Text Books

1. Tai-Ran, H. (2002). MEMS & microsystems: design and manufacture. ME Series.
2. Madou, M. J. (2002). Fundamentals of Micro-fabrication: the science of miniaturization. CRC

Reference Books:

1. Campbell, S. A. (2001). The Science and Engineering of Microelectronic Fabrication (The Oxford Series in Electrical and Computer Engineering).

Online Course Material:

- MEMS and Microsystems, IIT Kharagpur, Prof. Santiram Kal https://nptel.ac.in/courses/117105082/

Title	**Additive Manufacturing**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [E]	3–0–0 [3]
Offered for	B.Tech/M.Tech.	Type	E
Prerequisite	Nil		

Objectives

The Instructor will:

1. cover scientific as well as technological aspects of various additive manufacturing processes.
2. cover wide range of contemporary methodologies/technologies and tools for rapid manufacturing.

Learning Outcomes

The students are expected to have the ability to:

1. appreciate process flow and working principles for various additive manufacturing processes
2. deal with challenges faced during additive manufacturing of components

Contents

1. Introduction: Introduction, historical perspective, Classification of processes, AM enabling technologies, process chain for AM, sustainability perspective (5 Lectures)
2. Data Preparation: Data formats, data conversion, data validity checks, data repair (3 Lectures)
3. Process Planning Stage: Slicing Algorithms, Part Depositions and Orientation, Direct Slicing, computer aided process planning for AM (4 Lectures)
4. AM Processes: Introduction, machine architecture, process simulation, defects, advantages and disadvantages for various processes - Vat photopolymerization (5 lectures), Material jetting and Binder jetting (3 lectures), Laminated Object manufacturing (1 lecture), Extrusion based processes (3 lectures), Powder bed fusion processes (4 lectures), Direct energy deposition (2 lectures)
5. Design for Additive manufacturing (3 Lectures)
6. Process simulations of Additive Manufacturing processes (3 Lectures)
7. Additive manufacturing in context of Industry 4.0 (1 Lecture)
8. Process Selection and Industrial case studies (2 Lectures)

Text Books

1. Gibson, I, Rosen, D W, and Stucker, B (2021). Additive Manufacturing Methodologies: Rapid Prototyping to Direct Digital Manufacturing, Springer.
2. Leary, M (2020). Design for Additive Manufacturing, Elsevier

Reference Books

1. Hopkinson, N, Haque, R, and Dickens, P (2005). Rapid Manufacturing: An Industrial Revolution for a DigitalAge, Wiley
2. Chua, C K, Leong, K F, Lim C S (2010). Rapid Prototyping, World Scientific

Online Course Material:

- Fundamentals of Additive Manufacturing Technologies – https://nptel.ac.in/courses/112103306

1.

Title	**Theory of Arc Welding**	Course Code	MEL7XX0
Department	Mechanical Engineering	L-T-P [E]	3-0-0-0 [3]
Offered for	PG	Type	E
Pre-requisite	Nil		

Objectives

1. To impart knowledge and train students in area of arc welding processes

Learning Outcomes

1. Acquire fundamental understanding of the principles of arc welding
2. Develop detailed understanding of Arc welding processes and effect of their process parameters
3. Understand how arc welding processes are selected, controlled, and applied to the various joint design requirements and Diagnose faults in arc welding processes and acquire basic understanding of the heat flow and residual stresses in welds

Course Content

Physics of welding arc:- Welding arc, voltage distribution along the arc, thermionic and non-thermionic cathodes, arc characteristics and its relationship with power source, arc efficiency, heat generation, effect of shielding gases on arc, isotherms of arcs, Classification, forces acting on the drop, metal transfer mechanisms, transition current, melting rate, effect of polarity, deposition efficiency, current and voltage oscillograms, high speed films. (10 Lectures)

Consumable electrode welding processes:- Manual metal arc (MMA) welding: type composition and functions of flux covering, ISI and other international codes for electrodes, concepts of special electrodes, consumables, arc length control in pulsed MIG welding, selection of parameters, self-shielded and gas shielded flux cored wire welding.(6 Lectures)

Submerged Arc Welding: Specific features, process variables, types and composition of fluxes and their manufacturing, arc length control, significance of flux-metal combination; Non-Consumable Electrode Welding Processes:- Gas tungsten arc welding, electrodes, compositions, shielding gases, arc ignition and maintenance, selection of polarity, arc voltage rectification and remedy, cathode spot and normal mode operations. Plasma arc welding: transferred and non-transferred plasma arc welding, selection of gases, welding parameters, keyhole technique. (10 Lectures)

Heat flow and Residual Stresses in Welds:- Heat flow in welding, effect of welding parameters on heat distribution, Calculation of peak temperature, weld thermal cycle, cooling rate and solidification time, residual stress distribution, influence of residual stress in static and dynamic loading. (14 Lectures)

Reference Books

1. Messler R. W., (1999), Principles of Welding (Processes, Physics, Chemistry and Metallurgy), John Wiley

Reference Books

2. Metals Handbook, Vol. 6, (1993), ASM International Publication
3. Welding Handbook Vol. 1, 2 & 3, (1991), American Welding Society

Online Learning Material

- Welding Engineering, IIT Roorkee, Dr. D. K. Dwivedi http://nptel.ac.in/courses/112107089/

Title	**Precision and Micro Manufacturing**	Number	MEL7XX0

Department	Mechanical Engineering	L-T-P[E]	3–0–0 [3]
Offered for	PG	Type	E
Prerequisite	Nil		

Objectives

The instructor will :

1. Teach important topics in precision, micro/nano manufacturing based on the fundamentals enabling technologies.

Learning Outcomes

The students are expected to have the ability to:

1. understand the design and process issues associated with precision machine tools and the fabrication of precision components
2. develop a conceptual design solution to a precision machine operating in the small scale range
3. understand various fabrication techniques in manufacturing micrometer scaled devices and systems of complicated geometry

Contents:

1. Introduction to Precision engineering and practices: definitions, sources of error (4 Lectures)
2. Basic concepts of machining and precision engineering: Machine tool variables-accuracy, repeatability, stiffness, spindle vibration, flatness, straightness, and smoothness of motion, 1-2 DOF systems, feedback variables, cutting tool variables, workpiece variables, environment effects and thermal errors. Machine design for precision manufacturing, principles of measurement mechanical errors. (5 Lectures)
3. Micro machining techniques: Diamond micromachining introduction, diamond as a tool material, compatible materials, diamond performance, diamond machining, micro-mechanical applications. Micro-ECM, micro-EDM, Micromilling: micro-milling tools, process results and micro-milling applications. Microdrilling: micro-drilling and macro-drilling techniques. Laser micromachining: laser optics, laser ablation, heat affected zone and laser polymerization. (16 Lecture)
4. Joining Processes: Joining processes in similar and dissimilar materials; welding processes like ultrasonic, electron beam, laser beam and associated applications. (5 Lectures)
5. Micro casting: casting processes like vacuum, semi-solid state; applications processing of integrated circuits, clean rooms, crystal growing and shaping of wafers, etching, photo and other lithography techniques, impurity introduction, thermal oxidation, CVD etc. (12 Lectures)

Text Books

1. Murthy R. L., Precision engineering manufacturing, New Age International, 2005
2. Suryaprakash M. V., Precision Engineering, Narosa publications

Reference Books

3. Venkatesh V. C., Precision Engineering, McGraw Hill Publications
4. Nakazawa H., “Principles of precision engineering”, Oxford University Press

Self-Learning Material:

1. http://library1.org/_ads/A893BCE522D62C29FFB1D4CED8405ED4
2. https://nptel.ac.in/syllabus/112104028/

Title	**Computer Aided Inspection and Quality Control**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [E]	3-0-0-0
Offered for	PG	Type	E
Prerequisite	Nil		

Objectives:

The instructor will

1. introduce contact and non-contact techniques for the inspection of components
2. provide brief understanding about concepts of computer aided quality control

Learning Outcomes:

The students are expected to

1. appreciate fundamentals of contact and non-contact inspection techniques along with quality control tec for its implementation in various measurement applications

Contents

1. Contact Inspection Techniques: Fundamentals, Coordinate Measuring Machine, Programming Compensation, DMIS file components, Computer Aided Inspection Planning, Automatic Feature a extraction, Integration of CAD and Inspection Planning (14 lectures)
2. Non-contact Inspection Techniques: Introduction to Machine Vision and its applications, Image Acqu Digitization techniques, Image Proessing and Analysis Techniques, Interpretation of the image da inspection methods – Comparators, Laser Scanners and Linear Arrays, Non-optical methods – Techniques, Ultrasonic methods (14 lectures)
3. Computer Aided Quality Control: Quality in Design and Manufacturing, Traditional vs Computer Aic control, Process variability and capability, Statistical process control, Six Sigma and DMAIC, Taguchi Quality Engineering, International Standards (14 lectures)

Text Books

1. Al-Ahmari A., Nasr E. A., and Abdulhameed O., (2016), Computer Aided Inspection Planning: Theory and Practice, CRC Press

Reference Books

2. Demant C., Streicher-Abel B., and Waszkewitz P., Industrial Image Processing – Visual Quality Contro Manufacturing, Springer
3. Montgomery D., (2005), Introduction to Statistical Quality Control, John Wiley

Online Study Material:

- Computer Integrated Manufacturing, IIT Kanpur, Prof. J. Ramkumar

https://nptel.ac.in/courses/112104289

Title	**NDT for Manufacturing Applications**	Number	MEL7XXX
Department	Mechanical Engineering	L-T-P [E]	3–0–0 [3]
Offered for	UG & PG	Type	E
Prerequisite	Nil		

Objectives

The instructor will:

1. To introduce the fundamental principles of non-destructive testing and their importance in manufacturing quality control and process monitoring.
2. To develop an understanding of various NDT techniques and their suitability for different materials, manufacturing processes, and defect types.
3. To familiarize students with modern industrial practices, including inline, automated, and digital NDT systems used in manufacturing.

Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Explain the role of NDT in ensuring product quality and reliability across different manufacturing sectors.
2. Select appropriate NDT methods for inspecting castings, welds, forgings, and additively manufactured components.
3. Interpret inspection results and relate them to manufacturing processes for effective decision-making in industry.

Contents:

Module 1: Introduction to Non-Destructive Testing (10Lectures)

Definitions and scope: NDT, NDE, and SHM; Overview of standards and codes; Classification of NDT methods: Mechanical, Ultrasonic, Electromagnetic, Thermal, Radiographic, Optical.

Module 2: Ultrasonic and Electromagnetic NDT Techniques (16Lectures)

Basic Principles of Ultrasonics; Ultrasonic Wave-Material Interaction; Ultrasonic Transducers; Ultrasonic Testing Methods and Applications; Signal Processing and Imaging in UT; Electromagnetic Theory for NDT; Eddy Current Testing (ECT); Electromagnetic Acoustic Transducers (EMATs).

Module 3: NDT in Manufacturing Processes (13Lectures)

Inspection of Manufacturing Defects; Radiographic Testing Techniques; Thermal Non-Destructive Evaluation; Optical and Full-Field NDT Techniques; Inline and Automated NDT Systems; Selection and Reliability of NDT Methods; Emerging Trends in Manufacturing NDT.

Textbooks

1. J. Prasad and C. G. K. Nair, Non-Destructive Test and Evaluation of Materials, Tata McGraw-Hill Education, 2nd edition (2017).
2. J. L. Rose, *Ultrasonic Guided Waves in Solid Media*, Cambridge University Press.
3. Hirao, M. and Ogi, H., Development of EMAT Techniques in EMATS FOR SCIENCE AND INDUSTRY. Kluwer Academic Publishers, London, 13-82, (2004).
4. B. Raj, T. Jayakumar, and M. Thavasimuthu, Practical Non-Destructive Testing, Alpha Science International Limited, 3rd edition (2007).
5. Ed. Peter J. Shull, Non-destructive Evaluation: Theory, Techniques, and Applications, Marcel Dekker (2002).

Reference Books and Standards

1. C. Hellier, Handbook of Non-Destructive Evaluation, McGraw-Hill Professional, 1st edition (2001).
2. ASM Handbook, Volume 17: Non-destructive Evaluation of Materials, ASM International (2018).
3. Don E. Bray and Roderic K. Stanley, Non-destructive Evaluation: A Tool in Design, Manufacturing and Service, CRC Press (1996).

Online Course Material

1. https://www.nde-ed.org
2. NPTEL online course, 'Theory And Practice of Non-Destructive Testing,' Prof Ranjit Bauri, IIT Madras, https://onlinecourses.nptel.ac.in/noc20_mm07/preview

Title	**Engineering Design Innovation**	**Number**	MEL7620
Department	Mechanical Engineering	**L-T-P [E]**	3-0-0 [3]
Offered for	B. Tech (3rd year), M.Tech., Ph. D	**Type**	E
Prerequisite	Manufacturing Concepts		

Course Objectives

- Understand and utilize the engineering product design process as a framework for innovation.
- Learn fundamental concepts and applications of Design Thinking.
- Conceptualize, design, and prototype innovative solutions to real-world problems.
- Foster critical thinking and problem-solving skills through system design and system thinking methodologies.

Learning Outcomes

Upon completion of this course, students will be able to:

- Articulate and apply the Design Process in a business context.
- Conduct user research and empathize with diverse target audiences.
- Rapidly develop and test innovative ideas through iterative cycles.
- Create physical prototypes and visual representations to communicate concepts.

CONTENTS

Understanding Design

Introduction to Design Principles and History, Science, Engineering, and Psychology in Design, Product Lifecycle and Systematic Design Process, Technology's Role in Design [Lectures 6]

Thinking Methods

Designer's Mindset and Overcoming Creative Blocks, Critical, Creative, and Lateral Thinking, Introduction to Design Thinking, Design Thinking frameworks, and Stages of Design Thinking [Lectures 12]

Engineering Product Design and Development

Characteristics of Successful Products, Product Design and Development Processes [Lectures 3]

Problem Design

Opportunity Identification and Customer Needs, Task Clarification and User Research, Problem Identification and Analysis [Lectures 10]

Conceptual Design

Introduction to Conceptual Design, Imagination, Motivation, and Inspiration, Ideation Techniques, Concept Generation and Visualization Tools, Concept Evaluation and Selection [Lectures 8]

TEXTBOOKS

1. Pahl, G., Beitz, W., "Engineering Design: A Systematic Approach," Vol. 3, Springer, Germany, 2007.
2. Ulrich, K., Eppinger, S., "Product Design and Development," Vol. 7, McGraw-Hill/Irwin, USA, 2020.

REFERENCES

1. Brown, Tim, "Change by Design," Harper Collins Publisher, USA, 2020.
2. Dyson, James, "Against the Odds: An Autobiography," Texere Publishing, USA, 2003.
3. Lewrick, Michael, "Design Thinking and Innovation Metrics," Wiley Publications, USA, 2023.
4. Eriksson, Yvonne, "Different Perspectives in Design Thinking," CRC Press, USA, 2022.

Online Resources: https://onlinecourses.nptel.ac.in/noc21_me66/preview
https://onlinecourses.nptel.ac.in/noc22_me17/preview

Title	**Welding: Metallurgy, Defects and**	Number	MEL7XX0

	Testing		
Department	Mechanical Engineering	L-T-P [E]	3-0-0-0 [3]
Offered for	PG	Type	E
Prerequisite	Nil		

Objectives:
The aim of the course is to provide fundamental principles of metallurgy related with welding. Knowledge about the welding defects and their testing.

Learning Outcomes:
1. Understanding about welding Metallurgy
2. Imparting knowledge about the welding defects and factors responsible for it.
3. Role of diffusible hydrogen content and residual stresses on mechanical performance of the welded joint.
4. Knowledge about the non-destructive testing of welded joint.
5. Imparting knowledge about the heat treatment of the welded joint.

Contents:
1. Fundamentals of physical metallurgy: Need, phase diagrams: Fe-C, Al-Cu, Cu-Zn system, Phase transformations in Fe-C system, TTT diagram and CCT diagram, Carbon equivalent, Schaffer diagram, relevance of above in welding, delta ferrite formation (8 Lectures)
2. Effect of welding parameters on heat distribution, calculation of peak temperature, Residual stresses and shrinkage stresses, influence of stress in static and dynamic loading (6 Lectures)
3. Welding defects (Hot cracking, cold cracking, solidification cracking, nonmetallic inclusions; lamellar tearing; hydrogen damage) and non-destructive testing (Visual inspection, Liquid penetrant, Magnetic particle, Eddy current, X-ray, Radiographic, Ultrasonic testing) (10 Lectures)
4. Diffusible hydrogen content in deposited metal their measurement, Role of diffusible hydrogen on hydrogen induced cracking (2 Lectures)
5. Metal strengthen approaches: introduction, solid solution strengthening, grain refinement, precipitation hardening, transformation hardening, dispersion hardening, work hardening, strain aging (4 Lecture)
6. Heat treatment of weld joint: Need, Annealing; Normalizing; Quenching; Tempering; Austempering; Martempering and stress relieving of steel, Precipitation hardening of Al and copper alloys, Solidification of weld metal: principle of solidification of weld metal, modes of solidification, effect of welding parameter on weld structure, grain refinement principle of weld metal, method of weld metal refinement (10 Lecture)

Text Books:
1. Lancaster J F., Metallurgy of Welding, Allen & Unwin

Reference Books:
2. Avner S D, Introduction to physical metallurgy, TMH
3. ASM Handbook Vol. 6, Welding, Brazing and soldering, ASM International
4. Kou S., Welding metallurgy, 2nd edition, Wiley

Online Study Material:
- Fundamental of Welding Science and Technology by Prof Pankaj IIT Guwahati, https://www.youtube.com/playlist?list=PLwdnzlV3ogoUQnGO8eFFygVBTjF0xyYMq

Title	Vacuum Technology in Manufacturing	Number	ME7XX0
Department	Mechanical Engineering	L-T-P [E]	3–0–0 [3]
Offered for	B. Tech./ M. Tech.	Type	Elective
Prerequisite	Nil		

Objectives

Upon successful completion of this course, students will be able to:

1. Understand the fundamental principles of vacuum science, vacuum regimes and their relevance to manufacturing.
2. Design a vacuum system by considering mechanical, material, and vacuum constraints and analyse the performance.
3. Relate the vacuum technology requirements to advanced manufacturing processes.

Learning Outcomes

Students are expected to:

1. Understand the vacuum fundamentals, analyse gas loads, flow regimes, and pumping performance using quantitative methods.
2. Gain hands-on experience on vacuum system design, component selection, system testing, and troubleshooting using standard techniques.
3. Apply vacuum technology concepts to real-world manufacturing processes.

Content

1. Definition and importance of vacuum in manufacturing, Units of vacuum, Vacuum ranges, Gas laws, Knudsen number, Flow regimes, Gas sources in vacuum systems, Principles of vacuum pumping, Pumps classifications, pumping speed, Throughput, and ultimate pressure, Pump selection criteria, Vacuum symbols, Vacuum chamber design, Pumping system design, ASME Code for vacuum system design. [13 hours]
2. Vacuum valves, Vacuum measurement, Measurement ranges, Electrical, motion, and fluid feedthroughs, Viewports, Vacuum-compatible materials and lubricants, Types of leaks, Leak rate definition and acceptable limits, Helium mass spectrometer leak detection, Pressure rise method, Pump-down curves and system evaluation, Troubleshooting and safety practices. [13 hours]
3. Vacuum-based manufacturing applications, Vacuum system in electron beam welding and additive manufacturing, Vacuum heat treatment and brazing, Thin-film deposition and PVD processes, Plasma-based manufacturing processes, Vacuum casting and impregnation, Vacuum insulation in cryogenic systems, Vacuum freeze drying, Industrial and research case studies. [13 hours]

Text Books

1. VV Rao, KL Chopra & TB Ghosh, “Vacuum Science & Technology”, 1998.
2. Dr. Walter Umrath, “Fundamentals of Vacuum Technology.” Leybold Vacuum Handbook, New Edition 2016.

Reference Books

1. Jousten, K. (Ed.), “Handbook of vacuum technology.” John Wiley & Sons. (2016).
2. Farr JR. “ASME-BPVC Section VIII handbook for the design of pressure vessels.” ASME Press, New York, 2005.

Online Course Material

1. https://nptel.ac.in/courses/127105231

Review and Comments from Dean/EPC/SPGC/Senate Committee Meeting

1. **15 seats (8 MoE + 7 self-sponsored), DRDO can be extra**

   Response: As per suggestion of the committee, the strength has been modified.

2. **Rethink name of the program: M.Tech. in Manufacturing engineering**

   Response: The program name is revised as **"**M. tech in **Advanced Manufacturing Engineering"**

3. **Increase compulsory course to 4**

   Response:

   As per suggestion of the committee, a new compulsory course has been added to the revised course curriculum.

   Proposed New course: ***"Industrial and Manufacturing Systems Engineering"***

4. **Change name of 'Advances in manufacturing' to "Advanced Manufacturing Processes"**

   Response:

   The title of the course has been revised to ***"Conventional and Advanced Manufacturing Processes"***. The same has been suggested also by a few NRB experts.

5. **Add Microstructure and inspection related courses**

   Response:

   Microstructure related aspects have been added in course ***"Materials Design and Evaluation Techniques"***

   One dedicated course on inspection and quality control is already there in elective list, namely, "***Computer-Aided Inspection and Quality Control"***

6. **NDT should be there is compulsory course as few lectures**

   Response:

As per the suggestion of the committee, modification has been made and following content has been added in the program compulsory course *"**Materials Design and Evaluation Techniques"***

"Non-Destructive Testing (10 Lectures): Visual Inspection and Liquid Penetrant Inspection, Eddy Current Testing, Magnetic Particle Inspection, Ultrasonic Inspection, Radiography Inspection, Neutron Radiography, Case studies."

7. **Include AI/ML in manufacturing applications in elective course**

Response:

As per the suggestion of the committee, a new elective course ***"Artificial Intelligence for Manufacturing Applications"*** has been introduced in the revised course structure.

8. **Add aspects related with Green Manufacturing, Bio-manufacturing, and Automation and Industry 4.0/5.0**

Response:

As per the suggestion of the committee, the aspects related to Green Manufacturing and Bio-manufacturing has been added in the course content of ***"Conventional and Advanced Manufacturing Processes"*** (Program compulsory course). As per the suggestion of the committee, the aspects related to Automation and Industry 4.0/5.0 has been added in the course content of ***"Industrial and Manufacturing Systems Engineering"*** (Program compulsory course).

9. **The total program elective must be 3 times of offered elective courses.**

Response:

As per the suggestion of the committee, total 9 program elective courses has been proposed in the revised version of program.

Review and Comments on each course contents by International and National Advisory Board (NAB) Members

Reviewer 1

Prof. Rahul Mulik Professor, Mechanical and Industrial Engineering Department, IIT Roorkee		
S. N.	Comment	Response
1	The objectives are very clear and well written considering needs of the industries.	Thank you for the feedback.
2	The addition of hybrid manufacturing, MEMS, precision and micro manufacturing, and additive manufacturing etc is very interesting and much needed.	Thank you for the feedback.
3	Learning outcomes are also well defined.	Thank you for the feedback.
4	In case of electives, courses related to Sensors, Material characterizations may be added (you may refer, https://dm.iisc.ac.in/dm/#efaq-m-des-in-product-design-dev-management https://acad.iitr.ac.in/Varsity/Academic_Programmes/PG_2024/M.Tech.%20Additive%20and%20Joining%20Technologies.pdf etc.).	The aspects related to sensors are incorporated in "MEMS & Microsystems Technology," "NDT for Manufacturing Applications," and "AI for Manufacturing Applications" courses. The material characterization aspects are covered under the "Surface Engineering" course and "Materials Design and Evaluation Techniques."
5	The range of credits may be provided instead of fixed credit numbers (56).	As per the institute norms
6	2 Credit seminar for the students may be added so that they can get hands on experience related to literature review.	As per the institute norms
7	Internship option (with flexibility to join industries or academia) etc.	During the summer vacation, students can undertake internships to gain practical exposure and industry experience.

Reviewer 2

Prof. V. K. Jain Professor (Retired), Mechanical Engineering Department, IIT Kanpur		
S.N.	Comment	Response
1	Your course title is "Advanced Manufacturing Engineering" but your course curriculum has only a couple of courses in the name of "advanced manufacturing processes or systems".	Out of the four compulsory domain courses, two focus on advanced manufacturing engineering aspects, namely "Conventional and Advanced Manufacturing Processes" and "Advanced Material Processing." The

		remaining two address conventional manufacturing aspects, in line with the emphasis placed by other NAB members on including conventional manufacturing courses within the compulsory curriculum.
2	Valid GATE score in Mechanical Engineering/Aerospace Engineering/ Production and Industrial Engineering, or from any of the CFTIs with CPI or CGPA 8/10 or above. Comment: Valid Gate Score = 8/10 seems to be on the higher side. Make it 7/10.	As per the institute norms
3	Advanced Materials Processing 3-0-0 (3) can be made as an elective	The proposed course covers topics such as Introduction to Novel Materials, Advanced Powder-based Processing, Advanced Solid-state Processing, and Non-conventional and Hybrid Processing Techniques. The Manufacturing group feels that this course would be more appropriate if offered under the program compulsory category.
4	Additive Manufacturing, Precision and Micro-Manufacturing, Computer-Aided Inspection and Quality Control and Artificial Intelligence for manufacturing applications. These courses should be taught in 2nd or 3rd semester.	Will be prioritized while offering the electives.
5	Materials Evaluation Design can be removed as a compulsory course. It can be elective.	Another expert has also expressed support for retaining this course as a compulsory component of the program. The Manufacturing group respectfully concurs with and endorses this view.
6	You can add "Micromanufacturing Processes" and "Nano-finishing Science and Technology" courses if you feel appropriate. You can see the contents in the books edited by me and published by Taylor and Francis, New York, USA	The micro-manufacturing aspects are already covered in the proposed elective course "Precision and Micro Manufacturing." In addition, a separate elective course titled "Nano-finishing Science and Technology" has been included in the elective list. The suggested book has been considered as the textbook for this course.

Reviewer 3

Prof. Somashekhar S Hiremath, Professor, Department of Mechanical Engineering, IIT Madras		
S.N.	Comment	Response
1	Emerging technologies shaping engineering education such as Artificial Intelligence / Machine Learning / Data Science, Additive Manufacturing, Sustainable Manufacturing, Characterization Techniques, Surface Engineering of Advanced Materials for various applications. These topics are often offered only as elective courses, leading to limited exposure and	The topics such as Artificial Intelligence / Machine Learning are covered in the course "AI for Manufacturing Applications." The topic Sustainable Manufacturing is covered in "Conventional and Advanced Manufacturing Processes". Additive Manufacturing and Surface Engineering of Advanced Materials are addressed in "Advanced Material Processing."

	weaker conceptual grounding among many students. A small revision of core courses to integrate these areas could significantly strengthen the curriculum.	Characterization Techniques are covered in "Materials Design and Evaluation Techniques." In addition, a dedicated elective course on "Surface Engineering" has also been included in the curriculum.
2	For each subject, recommend 1–2 primary textbooks that cover at least ~60% of the syllabus.	The mentioned textbook adequately covers a major portion of the course syllabus.

Reviewer 4

Prof. Manas Das, Professor, Department of Mechanical Engineering, IIT Guwahati		
S. N.	Comment	Response
1	The curriculum of the proposed Advanced Manufacturing Engineering (AME) course effectively integrates theoretical knowledge with practical applications, enabling students to understand advanced manufacturing processes, automation, precision engineering, and emerging industrial trends. One of the major strengths of the program is its comprehensive coverage of both conventional and non-conventional manufacturing methods, along with modern tools such as CAD/CAM, additive manufacturing. The course structure promotes analytical thinking and problem-solving skills through projects, laboratory work, and research-oriented assignments. The course also prepares students for careers in core manufacturing industries, research organizations, and advanced production systems by emphasizing quality control, smart manufacturing, and sustainable engineering practices.	Thank you for the feedback.
2	Inclusion of more industry-driven case studies and live industrial projects	Projects addressing industry-relevant problems will be encouraged through term papers in the course, and case studies form an integral part of the teaching practices.
3	Greater exposure to Industry 4.0 tools, such as digital twins, AI in manufacturing, smart factories, and robotics, may be considered.	The topics such as Industry 4.0 and Digital Twins are already incorporated in the course "AI for Manufacturing Applications." The aspects related to smart factories is included in the curriculum of the same course.
4	More consideration should be given to advanced simulation software and modern manufacturing analytics.	Incorporated in the course "Industrial and Manufacturing Systems Engineering." Additionally, process simulation of additive manufacturing is already included in the "Additive Manufacturing" course.
5	Increased collaboration with industries for internships and practical training	Summer internship will be encouraged during the summer vacations

6	One core course, "Engineering Materials and Characterization," may be included in the syllabus.	Covered in the course content of "Advanced Material Processing" and "Materials Design and Evaluation Techniques."
7	Also, one lab core course (i.e., a manufacturing laboratory) may be included to gain hands-on experience with all the characterization facilities used in manufacturing.	Projects related to industry relevant problems are encouraged through term papers as part of the course.
8	Please rename the course name to Advanced Manufacturing Processes. Instead of the theory of arc welding, it may be an introduction to welding. Applied ergonomics may be removed from the pool of elective courses.	The title of the course "Advanced Manufacturing Processes" has been revised to "Conventional and Advanced Manufacturing Processes" to ensure inclusion of all conventional manufacturing processes. The title "Theory of Arc Welding" is retained to highlight the fundamental scientific principles underlying arc welding processes, while the course content provides an introductory overview. "Applied Ergonomics" is included to enable manufacturing engineers to incorporate human factors into system design, thereby enhancing workplace safety, operator efficiency, and overall manufacturing productivity. Another NAB member has also supported retaining this subject in the curriculum as an elective.
9	Overall, the AME program will be highly beneficial for students aiming to build expertise in advanced manufacturing technologies and research.	Thank you for the feedback

Reviewer 5

Prof. Ramesh Singh, Professor, Department of Mechanical Engineering, IIT Bombay		
S.N	Comment	Response
1	Engineering Mathematics: The course content looks good. The only thing would be to have some computational component to it. They may be exposed to Mathematica, Maple or any of the open-source symbolic maths software. This may be good benefit the students as they can solve complex problems using these tools.	The suggested comments will be forwarded to the respective course instructors to ensure that these suggestions are duly considered during the course instruction.
2	Materials Design and Evaluation Techniques: This is an interesting course and very useful for the manufacturing students to understand materials testing and selection. I would just recommend addition of these two topics by cutting the 12 lectures in Materials selection and Design to may be 8 lectures in 4 lectures	Theories of Failure" have been incorporated into the course curriculum of "Materials Design and Evaluation Techniques."

	one could cover Theories of Static and Dynamic Failure (Tresca, vonMises and Maximum Principal Stress) and some basic fatigue failure theories (Soderberg and Modified Goodman, the ASME and Gerber could be skipped).	
3	Advanced Materials Processing: Looks very packed with lot of information. Not sure if things can be covered in reasonable detail. This may be focused either on bulk manufacturing or surface engineering. So that the material could be covered in some detail.	Based on the suggestions, the course curriculum has been reviewed and certain topics have been streamlined to ensure that the content can be delivered with adequate depth. The revised curriculum primarily focuses on bulk manufacturing, while a dedicated elective course on "Surface Engineering" is already included to address related aspects.
4	Advanced Manufacturing Processes: The problem is that there is no machining and polymer processing. The conventional manufacturing processes have been completely ignored except for casting. In my opinion, this should cover the analysis of the conventional manufacturing processes, such as machining, grinding, bulk deformation, Sheet Metal forming, casting, welding, injection molding etc. These processes are used in the industry. In fact, the advanced process being emf forming, cryogenic forming and beam welding are used only for special process. Bulk of the manufacturing is still carried out by conventional processes, which is not being covered. I think the way it is covered in Ghosh and Malik with proper analysis may be the right way to go in this course.	The title of the course has been revised to "Conventional and Advanced Manufacturing Processes" to incorporate all the conventional manufacturing processes (such as machining, casting, welding, and forming), as appropriately suggested.
5	Industrial and Manufacturing Systems Engineering: The Automation in Manufacturing part could be decoupled and may be an elective course with basic control theory, industrial automatic PLC, hydraulic and pneumatic systems could be offered. The automation component does not appear to be very rigorous. Looks mostly as introduction to these topics.	A separate elective course titled "Automation in Modern Manufacturing Systems" has been designed to comprehensively cover all the suggested topics.
6	The electives are well designed. There are quite a lot of Design and Ergonomic courses, but fewer advanced elective courses in both systems and processes side. It will be good to encourage faculty to offer more courses on the systems side.	Several elective courses, namely "Nano-finishing Science and Technology," "Automation in Modern Manufacturing Systems," "Surface Engineering," "Advances in Casting Technology," and "Processing of Composites" have been incorporated to address the suggestions.

Reviewer 6

Prof. Bishakh Bhattacharya, Professor, Department of Mechanical Engineering, IIT Kanpur		
S. N.	Comment	Response
1	This is a great effort.	Thank you for the feedback

2	You should have two more elective courses: (a) Smart Materials and Intelligent Product Design (you may follow my NPTEL course on this), (b) Composite Materials	The aspects related to Smart Materials are covered in the "Advanced Material Processing" course. The topics on Intelligent Product Design are included in the "Engineering Design Innovation" course, while the aspects related to Composite Materials are addressed in the "Processing of Composites" course. In addition, the NPTEL course "Smart Materials and Intelligent System Design" has been incorporated as an online reference in the "Engineering Design Innovation" course.
3	The course on Precision and Micro manufacturing should be revised topics like Tribology, Metrology, vibration and shock isolation, micro-nano positioning should be included	The aspects related to Tribology are covered in the "Surface Engineering" course. The topics related to metrology and micro-nano positioning are addressed in the courses "Computer Aided Inspection and Quality Control," "Precision and Micro Manufacturing," and "MEMS and Microsystems Technology."
4	All courses should follow the same template - apparently the course on engineering design and systematic embodiment design are using a different format. Can they not be clubbed in one course?	Thank you for the suggestion. However, the two courses address distinct learning objectives and complementary aspects of the design process. Engineering Design focuses on the broader principles and methodologies of design, while Systematic Embodiment Design emphasizes the detailed development and realization of design concepts. Hence, it is considered appropriate to retain them as separate courses to ensure adequate depth and clarity in both areas.
5	The course on advanced manufacturing process should focus only on the process not materials.	The material-related aspects included in the curriculum of the mentioned course are limited in scope.
6	There should be a course on Patent, IP generation and commercialization.	Thank you for the suggestion. It is proposed to organize a dedicated 1–2-day workshop on Patents, IP generation, and commercialization. In addition, the IPR Cell of the institute will continue to provide guidance and support to students in these areas.

Reviewer 7

Prof. Hirotsugu Ogi Professor, Graduate School of Engineering, Osaka University, Japan		
S. N.	Comment	Response
1	The proposed M.Tech curriculum in Advanced Manufacturing Engineering is well-structured, academically strong, and aligned with global Industry 4.0 developments to prepare engineers for modern industrial challenges.	Thank you for the feedback.

2	The curriculum emphasizes advanced fabrication and materials processing while equipping students with essential skills in inspection, structural integrity evaluation, and quality assurance for high-reliability engineering systems.	Thank you for the feedback.
3	The curriculum adopts a forward-looking approach by integrating AI-driven manufacturing, modern design methodologies, innovation practices, and human-centered sustainable manufacturing concepts.	Thank you for the feedback.
4	Overall, the program provides a balanced integration of fundamental manufacturing science and emerging technologies, preparing graduates for advanced research and smart manufacturing, while its successful implementation will require sufficient highly qualified faculty across the interdisciplinary areas.	Thank you for the feedback.
5	Based on this evaluation, I recommend that the academic committee approve this curriculum.	Thank you for the feedback.

Reviewer 8

Prof. Ching-Chung Yin Professor, Department of Mechanical Engineering, NYCU, Taiwan		
S. N.	Comment	Response
1	The proposal presents a sophisticated and well-integrated framework for a Specialized Postgraduate Degree Program in Manufacturing Engineering. Its structural strength lies in the strategic equilibrium between the eight core and elective modules, which provide essential disciplinary breadth, anchored by a rigorous thesis requirement that mandates significant technical mastery and independent research.	Thank you for the feedback.
2	Through a comparative analysis of leading international and domestic frameworks, the proposal identifies in-situ monitoring as a critical technological frontier. Advancing this capability not only reinforces the four stated missions but also establishes a clear pathway toward sustainability objectives, positioning the manufacturing system as a global leader in responsible innovation.	Thank you for the feedback.

Reviewer 9

Dr. Kanwer Singh Arora, Tata steel, Jamshedpur

S.N.	Comment	Response
1	You may consider including a course on conventional manufacturing technologies such as casting, machining, and forming.	A course titled "Conventional and Advanced Manufacturing Processes" has been designed to incorporate all the conventional manufacturing processes, such as machining, casting, welding, and forming.
2	A course on materials — for example *Metallurgy for Non-Metallurgists* — covering both steels and non-ferrous metals would be a valuable addition.	The suggested topics have been incorporated into the course curriculum of "Advanced Material Processing."
3	A preliminary introductory course on electronics and semiconductors.	The suggested topics have been incorporated into the course curriculum of "MEMS & Microsystems Technology."
4	A course on HR practices, as the younger generation is increasingly less inclined toward manufacturing roles, especially those involving shop-floor environments.	Thank you for the suggestion. Instead of introducing a separate course, a dedicated workshop on HR practices in manufacturing will be organized to address these aspects and create greater awareness among students about shop-floor environments and manufacturing careers.

Reviewer 10

Dr. Vishwas Puttige, CEO and Director, Amace Solutions		
S. N.	Comment	Response
1	Can more content be added for industrial automation systems?	A dedicated course titled "Automation in Modern Manufacturing Systems" has been incorporated into the curriculum.
2	Can advanced subjects such as additive manufacturing be made as a core subject and not an elective?	The mentioned topics have already been incorporated in the last module of the course "Advanced Material Processing."
3	Have we covered subjects such as design for additive manufacturing too?	The suggested topics have already been incorporated in the "Additive Manufacturing" course.
4	I see AI for manufacturing, can we add more content on IoT, industry 4.0 and data analytics for manufacturing?	The suggested topics have already been incorporated into the "AI for Manufacturing Applications" course.
5	Can some applications based exposure be given to students especially in the 3rd and 4th semesters making them more industry ready?	Projects addressing industry-relevant problems will be encouraged through term papers, and case studies are incorporated as an integral part of the teaching practices.